

Place the following variables in a classification table. Identify whether each variable is discrete or continuous. Then classify them as nominal, ordinal, interval, or ratio-scaled variables.

- i. Height of students
- ii. Marital status
- iii. Number of books owned
- iv. Type of mobile operating system
- v. Performance level (Excellent, Good, Average, Poor)
- vi. Monthly household expenditure
- vii. Temperature in Celsius
- viii. Number of daily website visitors

Answer:

Variable	Discrete / Continuous	Scale Type	Justification
Height of students	Continuous	Ratio	Height can take any real value and has a true zero; ratios are meaningful.
Marital status	Discrete	Nominal	Categories such as single, married, divorced; no inherent order.
Number of books owned	Discrete	Ratio	Count variable with a true zero; ratios are meaningful.
Type of mobile operating system	Discrete	Nominal	Categories such as Android, iOS; no ranking exists.
Performance level (Excellent, Good, Average, Poor)	Discrete	Ordinal	Categories have a clear order but differences are not measurable.
Monthly household expenditure	Continuous	Ratio	Measured in money; true zero exists and ratios are meaningful.
Temperature in Celsius	Continuous	Interval	Equal intervals exist but no true zero (0°C does not mean absence of temperature).
Number of daily website visitors	Discrete	Ratio	Count variable with true zero and meaningful comparison.

2. Place the following variables in a classification table. Identify whether each variable is **discrete or continuous**. Then classify them as **nominal, ordinal, interval, or ratio-scaled variables**.

- i. Weight of employees
- ii. Blood group
- iii. Number of children in a family
- iv. Brand preference of smartphones
- v. Education level (SSC, HSC, Bachelor's, Master's, PhD)
- vi. Daily income
- vii. Temperature in Fahrenheit
- viii. Number of branches of a bank

Answer:

Variable	Discrete / Continuous	Scale Type	Justification
Weight of employees	Continuous	Ratio	Can take any real value; true zero exists; ratios are meaningful.
Blood group	Discrete	Nominal	Categories (A, B, AB, O); no ranking.
Number of children in a family	Discrete	Ratio	Count variable; true zero exists; ratios meaningful.
Brand preference of smartphones	Discrete	Nominal	Categories (e.g., Samsung, Apple, Xiaomi); no order.
Education level	Discrete	Ordinal	Ordered categories; differences between levels not measurable.
Daily income	Continuous	Ratio	Measured in money; true zero; meaningful ratios.
Temperature in Fahrenheit	Continuous	Interval	Equal intervals but no true zero.
Number of branches of a bank	Discrete	Ratio	Count data; true zero; meaningful comparison.

3. The following table shows the estimated values (in million dollars) of 28 professional football teams as reported by a sports business magazine.

950, 1100, 875, 1420, 1600, 2100, 1800,
990, 1200, 1340, 1560, 1700, 2200, 2400,
800, 1025, 1180, 1500, 1650, 1950, 2050,
730, 890, 1250, 1380, 1750, 2600, 3000

Requirement

1. Determine the appropriate number of classes using the 2^k rule.
2. Construct a frequency distribution table.
3. Compute:
 - i. Cumulative Frequency (CF)
 - ii. Relative Frequency
 - iii. Percentage
4. Draw a **Frequency Polygon**.
5. Comment on the shape of the distribution.

Step 1: Determine the number of classes using the 2^k rule

The $2^k \geq n$ rule is often used to determine the number of classes:

- Sample size $n = 28$
- Solve $2^k \geq 28$

Check powers of 2:

$$2^4 = 16(\text{too small})$$

$$2^5 = 32(\text{sufficient})$$

So, the number of classes $k = 5$

Step 2: Determine class width

Class width formula:

$$\text{Class width} = \frac{\text{Max} - \text{Min}}{\text{Number of classes}}$$

- Maximum value = 3000
- Minimum value = 730

$$\text{Class width} = \frac{3000 - 730}{5} = \frac{2270}{5} \approx 454$$

We can round to **450**

Step 3: Determine class limits

- Start with **lower limit** = 700 (close to min 730, for simplicity)
- Each class width = 450

Classes:

1. 700 – 1149
2. 1150 – 1599
3. 1600 – 2049
4. 2050 – 2499
5. 2500 – 2949

Check: last class upper limit should include 3000 → we can extend last class to 2950 → include 3000

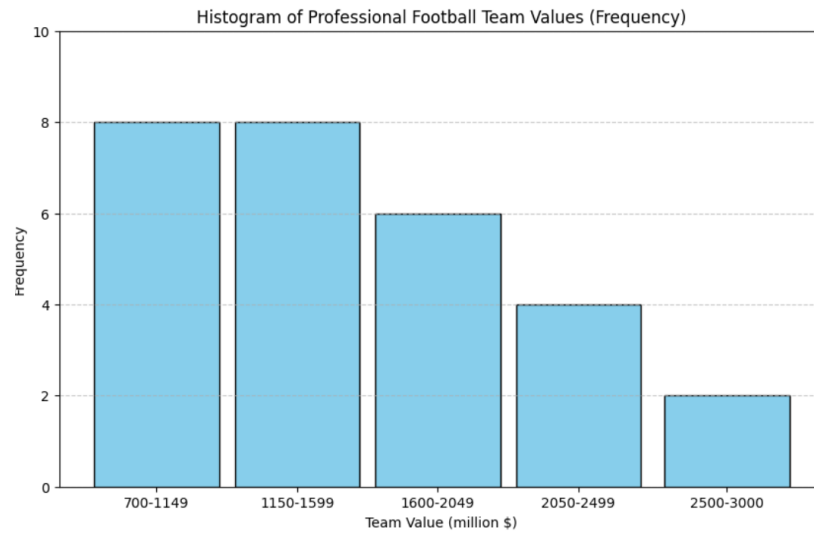
Step 4: Tally frequencies

Count how many values fall in each class:

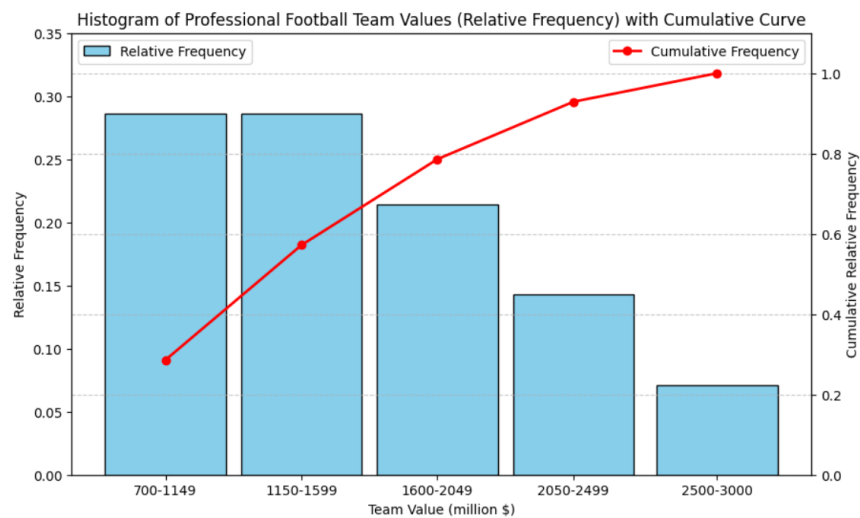
- i. **700 – 1149:** 950, 1100, 875, 990, 800, 1025, 730, 890 → 8
- ii. **1150 – 1599:** 1420, 1200, 1340, 1560, 1180, 1250, 1380, 1500 → 8
- iii. **1600 – 2049:** 1600, 1700, 1650, 1800, 1750, 1950 → 6
- iv. **2050 – 2499:** 2100, 2200, 2400, 2050 → 4
- v. **2500 – 3000:** 2600, 3000 → 2

Step 5: Frequency distribution table

Class Interval	Frequency	Cumulative Frequency	Relative Frequency	% Frequency
700 – 1149	8	8	0.286	28.6%
1150 – 1599	8	16	0.286	28.6%
1600 – 2049	6	22	0.214	21.4%
2050 – 2499	4	26	0.143	14.3%
2500 – 3000	2	28	0.071	7.1%
Total	28	-	1.00	100%



Here is the histogram showing the frequency of professional football team values. Each bar represents the number of teams in that value range.



Here's the histogram with relative frequencies (blue bars) and the cumulative frequency curve (red line).

- i. The bars show the proportion of teams in each value range.
- ii. The red line shows the running cumulative relative frequency, helping visualize how the data accumulates across classes.

4. Show the following in histogram and cumulative frequency polygon

Relative Frequency Distributions

- ▶ To find the relative frequencies, simply take the class frequency and divide by the total number of observations

TABLE 2-7 Relative Frequency Distribution of Profit for Vehicles Sold Last Month at Applewood Auto Group

Profit	Frequency	Relative Frequency	Found by
\$ 200 up to \$ 600	8	.044	8/180
600 up to 1,000	11	.061	11/180
1,000 up to 1,400	23	.128	23/180
1,400 up to 1,800	38	.211	38/180
1,800 up to 2,200	45	.250	45/180
2,200 up to 2,600	32	.178	32/180
2,600 up to 3,000	19	.106	19/180
3,000 up to 3,400	4	.022	4/180
Total	180	1.000	

▶ 2-12

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Graphic Presentation of a Frequency Distribution

HISTOGRAM A graph in which the classes are marked on the horizontal axis and the class frequencies on the vertical axis. The class frequencies are represented by the heights of the bars, and the bars are drawn adjacent to each other.

- ▶ A histogram shows the shape of a distribution.
- ▶ Each class is depicted as a rectangle, with the height of the bar representing the number in each class.

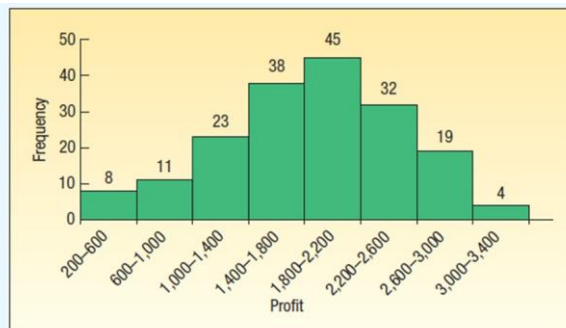


CHART 2-4 Histogram of the Profit on 180 Vehicles Sold at the Applewood Auto Group

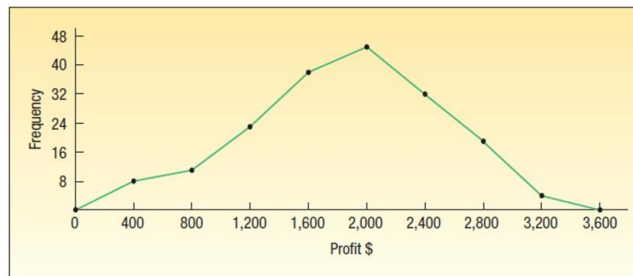
▶ 2-13

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Graphical Presentation of a Frequency Distribution

- ▶ A frequency polygon, similar to a histogram, also shows the shape of a distribution.
- ▶ These are good to use when comparing two or more distributions.

Profit	Midpoint	Frequency
\$ 200 up to \$ 600	\$ 400	8
600 up to 1,000	800	11
1,000 up to 1,400	1,200	23
1,400 up to 1,800	1,600	38
1,800 up to 2,200	2,000	45
2,200 up to 2,600	2,400	32
2,600 up to 3,000	2,800	19
3,000 up to 3,400	3,200	4
Total		180



Cumulative Frequency Distributions

Profit	Cumulative Frequency	Found by
Less than \$ 600	8	8
Less than 1,000	19	8 + 11
Less than 1,400	42	8 + 11 + 23
Less than 1,800	80	8 + 11 + 23 + 38
Less than 2,200	125	8 + 11 + 23 + 38 + 45
Less than 2,600	157	8 + 11 + 23 + 38 + 45 + 32
Less than 3,000	176	8 + 11 + 23 + 38 + 45 + 32 + 19
Less than 3,400	180	8 + 11 + 23 + 38 + 45 + 32 + 19 + 4

Cumulative Frequency Polygon

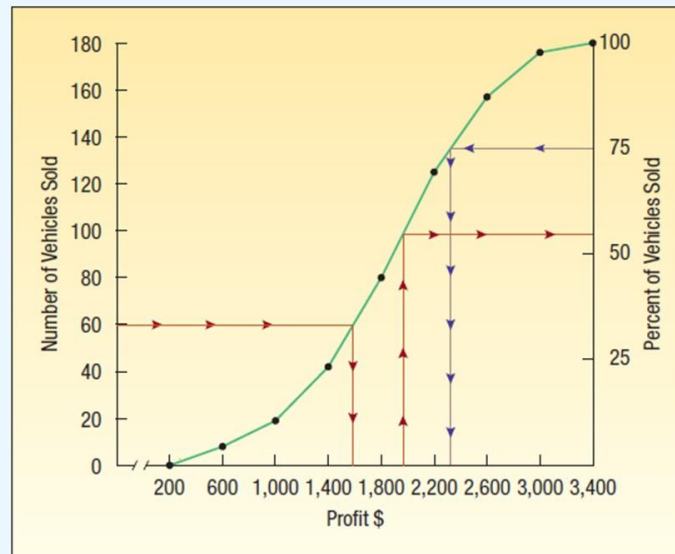


CHART 2-7 Cumulative Frequency Polygon for Profit on Vehicles Sold Last Month at Applewood Auto Group